ADULT ECHOCARDIOGRAPHY / A REFERENCE FOR FELLOWS AND CONSULTANTS

Echo Essentials

What the guidelines say, why they say it, and where readers go wrong.

- Chamber quantification and LV function
- Left atrial pressure, diastology and HFpEF
- The right heart and pulmonary hypertension
- Aortic stenosis, including both low-flow states
- Regurgitant lesions and mitral stenosis



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ABOUT THIS BOOK

Read it at the reporting station, not on the shelf.

Echocardiography guidelines are long, and the numbers you need are scattered across several of them. This book gathers those numbers in one place and keeps the reasoning attached to each one. Every chapter follows the same order: the criteria, the logic behind them, and the mistakes that catch out experienced readers.

It is written for cardiology fellows preparing for a lab rotation or an exam, and for consultants who want a quick, honest check before signing a report.

How the book handles disagreement

European and American documents agree on most thresholds and differ on a few. Where they differ, both values are printed and labelled. Where a number comes from one document only, the source is named in the table note.

Intervention thresholds in particular are moving. The early-intervention trials in asymptomatic aortic stenosis have already pushed practice ahead of older editions, so treat the decision tables here as a map of the landscape, and the guideline itself as the final authority.

Abbreviations used throughout

AVA, AVAi	Aortic valve area, indexed to body surface area
Vmax, MG	Peak aortic jet velocity, mean transvalvular gradient
DVI	Dimensionless velocity index, VTI_{LVOT} / VTI_{AV}
SVi	Stroke volume index, mL/m²
LFLG	Low-flow, low-gradient aortic stenosis
DSE	Dobutamine stress echocardiography
EROA, RVol, RF	Effective regurgitant orifice area, regurgitant volume, regurgitant fraction
LAP, LAVI	Left atrial pressure, left atrial volume index

SOURCES

ASE/EACVI chamber quantification (2015), valvular stenosis and regurgitation (2017), ASE diastolic function and HFpEF update (2025), ASE adult reporting standards (2025), 2025 ESC/EACTS valvular guidelines, ACC/AHA 2020.

PLEASE VERIFY

Classes of recommendation change between guideline editions. Before quoting one in a report or an exam, check it against the full current text.

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04

Aortic stenosis

Grade it, define it, and never miss the low-flow patient.

Aortic stenosis is the valve lesion where echocardiography most directly decides treatment, and also the one where small technical errors change the category. This chapter follows the order in which you work: acquire, grade, resolve discordance, then decide.

FOUR NUMBERS

Gradient, area, flow, ejection fraction. If these four do not tell one coherent story, one of the measurements is wrong.

4.1 Acquire three measurements, then add flow

- **Peak velocity.** Continuous-wave Doppler from several windows (apical, right parasternal, suprasternal) with a dedicated non-imaging probe. Report the highest clean signal and name the window.
- **Mean gradient.** Trace the same continuous-wave envelope. Do not mix gradients from different beats.
- **Valve area by continuity.** The LVOT diameter is squared in the calculation, so a 1 mm error shifts the area by roughly 0.2 cm^2. Measure just below the valve in mid-systole.
- **Dimensionless velocity index.** The ratio of LVOT to aortic VTI. It removes the LVOT diameter, and a value of 0.25 or below supports severe stenosis.
- **Stroke volume index.** LVOT area times LVOT VTI divided by body surface area. A value below 35 mL/m^2 defines low flow.

RULE

Poor alignment always underestimates velocity. It never overestimates it.

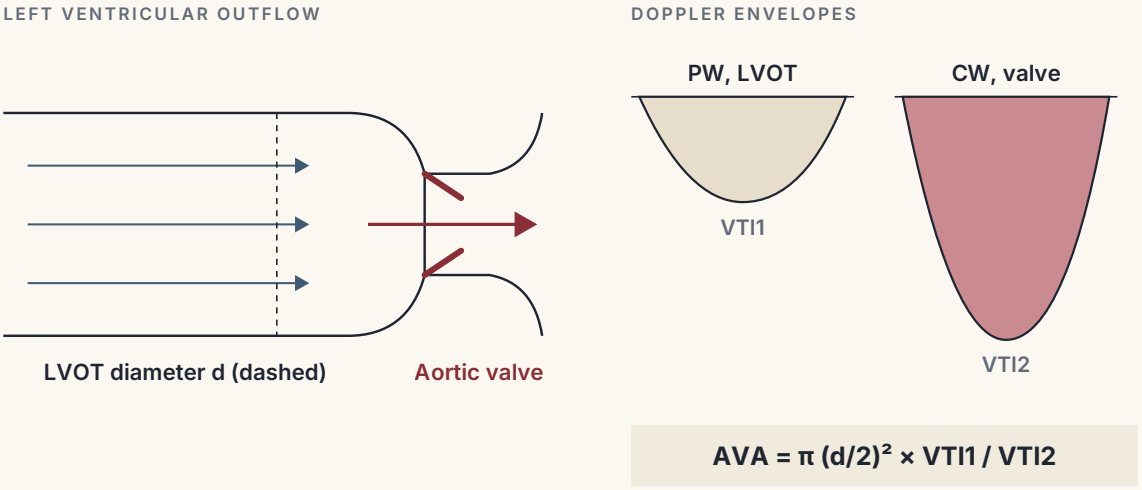


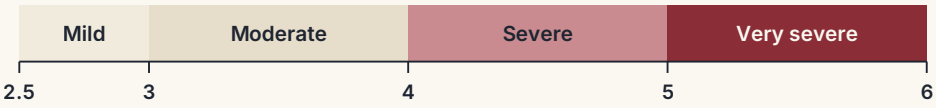
Fig. 4.1 The continuity equation. Flow below the valve (pulsed-wave, LVOT) equals flow through the valve (continuous-wave), so the unknown area can be calculated.

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4.2 Grade severity

Severe aortic stenosis is defined by a peak velocity of 4.0 m/s or more or a mean gradient of 40 mmHg or more, with a valve area of 1.0 cm^2 or less (indexed area 0.6 cm^2/m^2 or less). The ladder below places the other categories on the same axis.

PEAK AORTIC JET VELOCITY, Vmax (m/s)



Mean gradient, mmHg	< 20	20 to 39	40 to 59	≥ 60
AVA, cm^2	> 1.5	1.0 to 1.5	≤ 1.0	usually < 0.6
DVI	> 0.50	0.25 to 0.50	≤ 0.25	

Fig. 4.2 Severity ladder on the velocity axis. Severe: ESC/EACTS 2025, ASE/EACVI 2017 and ACC/AHA 2020 agree. Mild and moderate bands follow ACC/AHA 2020 staging.

CRITICAL AORTIC STENOSIS

In adults, critical (or very severe) stenosis means a peak velocity of 5.0 m/s or more and / or a mean gradient of 60 mmHg or more, usually with a valve area below 0.6 cm^2. ACC/AHA 2020 uses the term very severe. ESC/EACTS treats a velocity above 5.0 m/s as a high-risk marker that supports early intervention even without symptoms.

Be precise in the report. No society defines the word critical by one universal number. Give the measured values rather than the label alone.

In neonates, critical aortic stenosis is a different entity: a duct-dependent systemic circulation presenting in the first days of life. Adult cut-offs do not apply.

Mind the flow. A very high gradient needs adequate flow. A patient can have a critically narrow valve and a modest gradient when stroke volume is low.

4.3 When the numbers disagree

About one third of patients with a valve area of 1.0 cm^2 or less have a mean gradient below 40 mmHg. Do not average the two categories. Ask about flow first, then about the ejection fraction.

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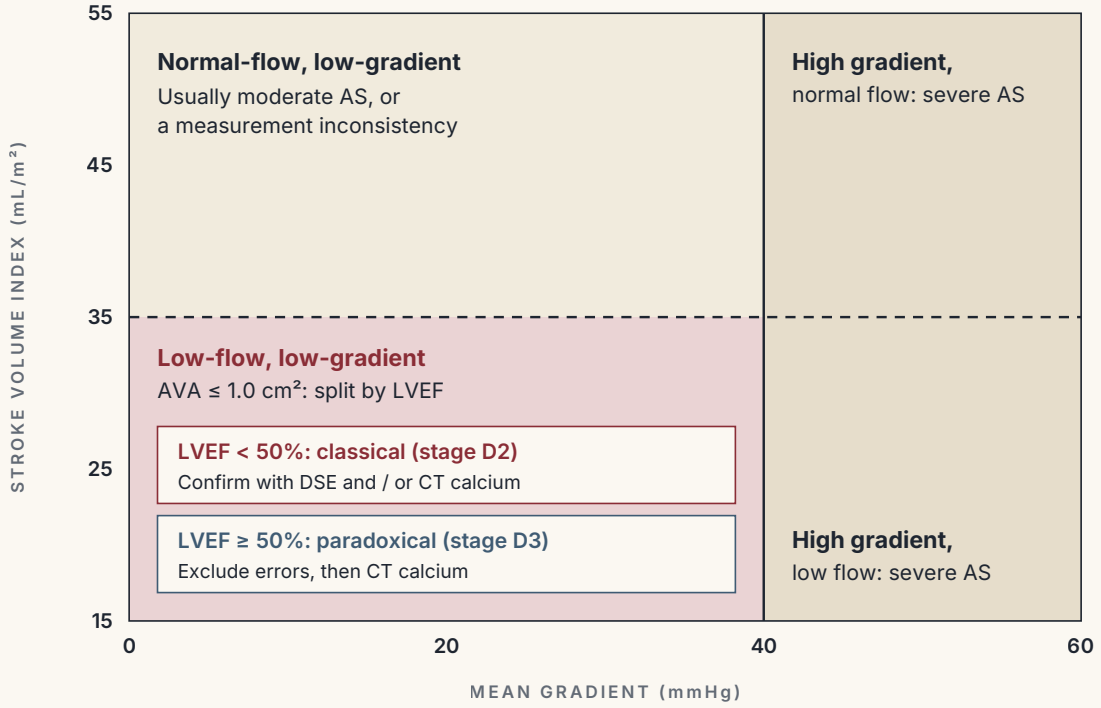
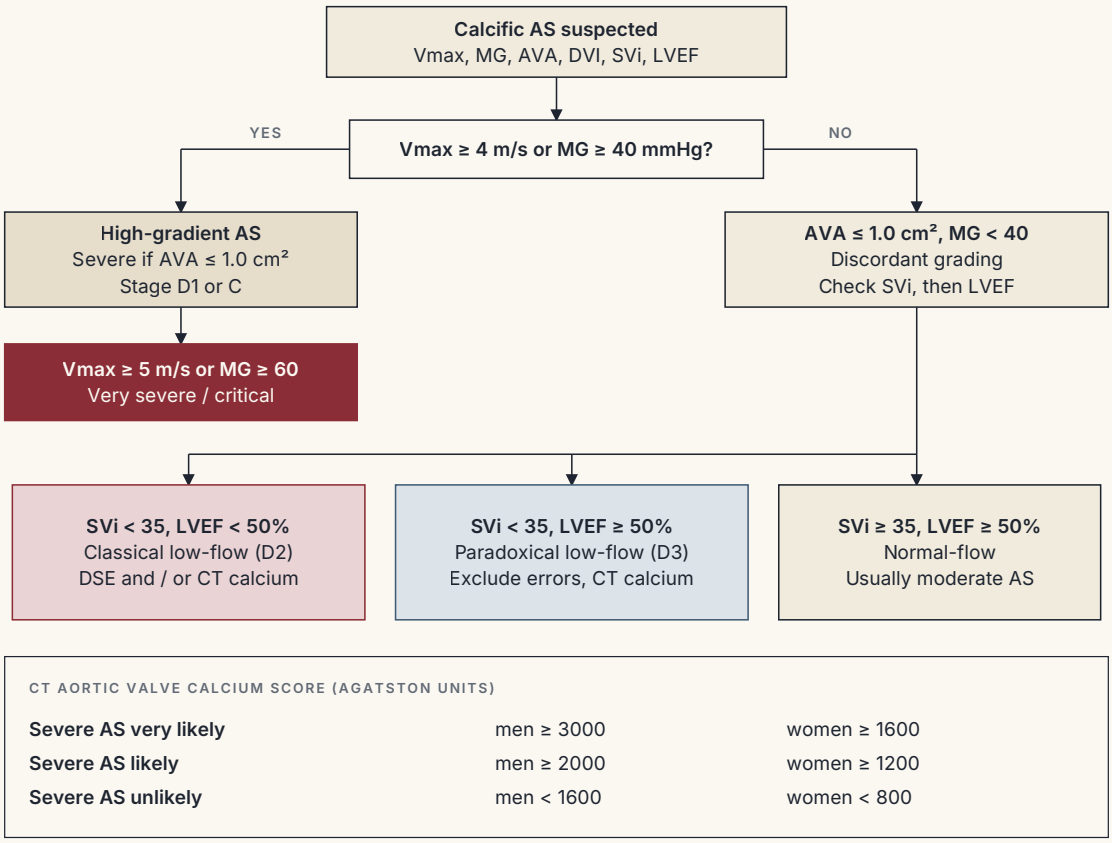


Fig. 4.3 A flow-gradient map of aortic stenosis. The shaded lower-left area is where the two low-flow states live. Both require confirmation, because both can be moderate stenosis in disguise.



CT AORTIC VALVE CALCIUM SCORE (AGATSTON UNITS)

Severe AS very likely	men ≥ 3000	women ≥ 1600
Severe AS likely	men ≥ 2000	women ≥ 1200
Severe AS unlikely	men < 1600	women < 800

Fig. 4.4 Working algorithm for discordant grading, with CT calcium thresholds. Stages follow ACC/AHA 2020. Calcium cut-offs follow ESC/EACTS (sex-specific).

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